Shubhadeep Mandal

81602 63797 shubhadeepmandal11@gmail.com shubhadeep-mandal Seronic2001

Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Projects

Peer-to-Peer Distributed File Sharing System | *C++, Multithreading, Sockets, OpenSSL, Linux*

- Built a BitTorrent-style file sharing system in C++ with a central tracker, multithreaded peers and per-piece integrity checks.
- Profiled and fixed bottlenecks in scheduling and networking, cutting a 1GB download from **193 s to under 13 s**.
- Added a replicated backup tracker with automatic failover, tested with unit tests under sanitizers.

FOOL: Custom POSIX-Style Interactive Shell | *C++, Linux, System Calls, Signals*

- Built a Unix shell in C++ directly on Linux system calls, with pipelines, redirection and job control.
- Fixed race conditions in signal handling so the shell never loses track of finished processes.
- Used `strace` to cut system calls in the line editor, and made startup **52x faster**.

Football Video Analytics: Video & Imaging Pipeline | *Python, OpenCV, H.264, YOLOv8*

- Built a video analysis pipeline that tracks players and the ball with a fine-tuned YOLOv8 model.
- Mapped the camera view to the pitch, correcting for camera motion, to measure player speed and distance.
- Streamed frames in batches and encoded video in the same pass, cutting memory use from **5.3 GB to 1 GB**.

Warehouse Traffic Control: Fault-Tolerant Robot Coordination | *JavaScript, Web Workers*

- Designed a lease-based protocol where 16 managers coordinate 200 robots with **zero collisions** over 416K moves.
- Added deadlock detection and crash recovery, so a failed manager rebuilds its state in seconds.
- Compared with a central coordinator: **7.6x lower** peak load per node, and a crash affects 11 robots instead of 120.

Technical Skills

Languages: C/C++, Python, Bash, SQL.
Linux: Processes, Signals, Pipes, Threads, Sockets.
Video: OpenCV, YOLOv8, H.264.
Tools: g++, Make, Git, strace, ASan/UBSan, Docker.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd by GPA in the B.Tech CSE batch at IIIT Ranchi.
- **PGEE 2025:** Secured All-India Rank 100 in the IIIT Hyderabad Post Graduate Entrance Examination.
- **Megathon 2025:** Placed 3rd in the mobile computing track.

Interests

- Reading web novels and manga.
- Playing chess and badminton.